1.6 Graphs of Other Trigonometric Functions

Name _____ Period ____ Date _____

Pre-Calculus 1.6 (30-minute plan) · Student sheet

I can…

- find the period and asymptotes of a tangent or cotangent function.
- sketch tangent and cotangent graphs.
- sketch secant and cosecant graphs using their reciprocal functions.

Vocabulary · fill in the missing word

Vertical asymptote — a vertical line x = a that a graph _____ but never reaches

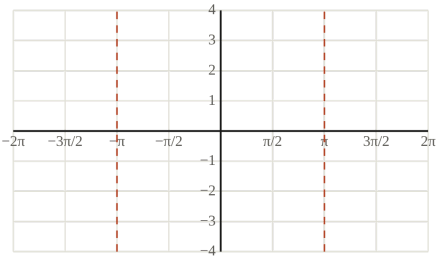
Reciprocal function — csc x = _____ = 1/cos x

Part 1 · Notes and examples (fill in each note, then work the example)

Tangent: y = a tan bx has period _____; asymptotes where bx = ±π/2.

Example 1

Sketch $y = \tan(\frac{x}{2})$. Find the period and two consecutive asymptotes.



Cotangent: y = a cot bx has period π/b; asymptotes where bx = 0, _____.

Example 2

Find the period and asymptotes of $y = 2\cot(\frac{x}{3})$ on one cycle.

Cosecant and secant: asymptotes where the sine (or cosine) is _____; the graphs touch at its maxima and minima.

Example 3

Find the period and the asymptotes on [0, 2π] of y = sec 2x.

Part 2 · Partner practice

Partner practice

Find the period of $y = 3 \csc 4x$.